

WJEC CBAC

Predicted Paper

GCE A LEVEL

1520U30-1

ECONOMICS – A2 unit 3

Exploring Economic Behaviour (Predicted Paper – Hard)

2 hours

Surname: _____ First name(s): _____

Centre Number: _____ Candidate Number: _____

For Examiner's use only

Question	Maximum Mark
Section A 1.	8
2.	7
3.	8
4.	6
5.	11
Section B 6.	10
7.	8
8.	10
9.	12
Total	80

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer all questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
You are reminded of the necessity for good English and orderly presentation in your answers.

SECTION A

Answer all questions in the spaces provided.

1.

Toyota is the world's largest carmaker by volume, producing over 10 million vehicles per year. Toyota's Tsutsumi plant in Japan produces more than 500,000 vehicles annually using specialised robotic assembly lines and the Toyota Production System (TPS), which is widely studied as a model of efficient manufacturing. Specialisation within the plant and long production runs help keep the unit cost of each vehicle low.

Toyota's presence in Aichi Prefecture has drawn in hundreds of component suppliers (known as 'keiretsu' firms), a dense network of local engineering schools, and transport infrastructure subsidised by the regional government. However, Toyota's global expansion has led to well-publicised difficulties: a 2010 recall of over 9 million vehicles was partly blamed on communication failures between distant divisions; middle-management layers expanded rapidly; and production mistakes at overseas plants have at times caused costly delays.

- (a)** Using a diagram and with reference to the data, outline the difference between internal economies of scale and internal diseconomies of scale. **[4]**

Draw your diagram in the space below.

- (b)** Using a diagram, outline how Toyota might benefit from external economies of scale in Aichi Prefecture. **[4]**

Draw your diagram in the space below.

2. The diagram below represents a profit-maximising firm operating in an oligopolistic market with a kinked demand curve.

[Diagram: kinked demand curve with price P^ on the y-axis. AR (demand) is elastic above the kink and inelastic below. MR has a vertical discontinuity at the kink. MC and AC curves shown, with MC passing through the vertical section of MR.]*

Key curves shown: MC, AC, AR (kinked demand), MR (with discontinuity at the kink).

- (a)** On the diagram, indicate the profit-maximising level of output produced by this firm. **[1]**
- (b)** Shade the area that represents the firm's total fixed costs of production. **[2]**
- (c)** Explain why the firm's price is likely to remain stable even if its marginal cost changes slightly. **[4]**

3.

Japan Post (Nippon Yūsei) is the giant state-owned postal, banking and life insurance conglomerate. At the time of its partial privatisation in 2015, it held over ¥200 trillion (US\$1.8tn) in deposits and insurance funds, making it one of the largest financial institutions in the world. Japan Post was privatised in stages, with the government initially selling around 11% of shares and continuing to hold a majority stake.

The reform aimed to improve efficiency, release capital for investment, and allow Japan Post to compete with private banks and insurers on equal terms. Supporters argued Japan Post's deposits were being inefficiently recycled into low-return Japanese Government Bonds, crowding out private investment. Critics countered that Japan Post's network of 24,000 post offices served an ageing rural population who depend on its savings and insurance products; privatisation might lead to branch closures and loss of universal service.

Post-privatisation, Japan Post's acquisition of Australia's Toll Holdings in 2015 resulted in a ¥400bn write-down. The company's share price has underperformed the Nikkei. Rural branch closures have been limited by regulation, but critics say service quality has declined. In 2021, mis-selling scandals at Japan Post Insurance prompted further reform pressure.

Evaluate the view that Japan's partial privatisation of Japan Post has been successful. [8]

4. The Bank of Japan (BoJ) ran negative interest rates from 2016 to 2024. Figure 1 shows the BoJ policy rate over this period.

Figure 1: Bank of Japan short-term policy rate, 2015–2024 (%)

Year	2015	2016	2018	2020	2022	2024 (Mar)	2024 (Jul)
Policy rate (%)	0.10	−0.10	−0.10	−0.10	−0.10	0.10	0.25

Analyse the likely impact of negative interest rates on aggregate demand in Japan. [6]

5. For over two decades, Japan experienced near-zero or negative inflation. In 2022–2023 Japanese inflation rose sharply due to a weak yen and rising global commodity prices. The BoJ has a 2% CPI inflation target.

Figure 1: Japan core CPI inflation rate 2018–2024 (annual % change)

Year	2018	2019	2020	2021	2022	2023	2024
Core CPI (%)	0.3	0.5	-0.4	-0.2	2.3	3.1	2.5

Figure 2: Japan annual inflation rate for selected product categories, April 2022 to April 2023

Category	Energy	Food (excl. fresh)	Durable goods	Services	Rent	Recreation
Annual inflation rate	14.6%	9.0%	7.4%	1.8%	0.4%	3.7%
Weighting (% of spending)	7	22	12	50	7	2

(a) Using the data in Figure 2, calculate a price index for Japan in April 2023, taking April 2022 as the base period. You are advised to show your working. **[3]**

Figure 3: Index of real and nominal average wages in Japan, 2019–2023 (2023 = 100)

Year	Nominal wage index	Real wage index
2019	98	104
2020	96	103
2021	97	103
2022	99	102
2023	100	100

In Japan's 'shuntō' (spring wage negotiations), 2023 delivered average wage rises of 3.6% — the largest in thirty years — and 2024 produced 5.1% rises. However, real wages remained negative for 21 consecutive months through 2024.

(b) Assess whether Japan is likely to experience a wage-price spiral. **[8]**

SECTION B

Answer all questions in the spaces provided.

Japan's yen crisis and 'Abenomics' reckoning

In 2022 the Japanese yen (¥) collapsed to a 32-year low against the US dollar, depreciating from ¥115/US\$ at the start of the year to over ¥150/US\$ by October. Although the yen recovered partially by 2024, it remained historically weak, touching ¥161/US\$ in July 2024 before the Bank of Japan finally ended its negative-interest-rate policy. Japan is the world's third-largest economy and its currency moves have global financial market implications.

Figure 1: USD/JPY exchange rate, 2021–2024 (yen per US dollar)

Date	Jan 2021	Jan 2022	Oct 2022	Jan 2023	Jul 2024	Oct 2024
¥ per US\$	103	115	150	131	161	152

(A higher number means a weaker yen.)

Japan is highly dependent on imports of energy and food. The country imports around 90% of its oil and over 60% of its food. Combined with rising global commodity prices from 2021 onwards, yen weakness created severe inflationary pressure for Japanese consumers and firms for the first time in a generation.

Table 1: Prices of selected Japanese imports 2020–2023 (¥/tonne or ¥/barrel)

Commodity	2020 price (¥)	2023 price (¥)
Crude oil (¥/barrel)	4,400	11,800
LNG (¥/tonne)	47,000	143,000
Wheat (¥/tonne)	27,000	54,000
Beef (¥/kg import)	760	1,320

Japanese firms have absorbed much of these cost increases, with profit margins squeezed. However, from 2023 onward, companies began passing costs through to consumers: in a society where price rises were long considered taboo, firms such as Yamazaki Baking and Kewpie mayonnaise increased prices repeatedly, to consumer outcry. Small businesses have been particularly vulnerable — Teikoku Databank recorded a 30% year-on-year rise in corporate bankruptcies in 2023, with yen weakness cited as a primary factor for import-dependent firms.

Japan's central bank faced a policy dilemma. Raising interest rates would support the yen and tame imported inflation, but risked triggering a 'carry-trade unwind' — global investors have borrowed yen cheaply for decades to invest in higher-yielding assets abroad, and a Japanese rate rise could cause sharp global market volatility (as was briefly seen in August 2024). Higher rates would also raise debt-service costs on Japan's government debt, which at 263% of GDP is the highest of any developed economy. Alternatively, the BoJ could intervene directly in currency markets by selling US dollars and buying yen.

Figure 2: Japan's foreign exchange reserves (US\$ billion, month-end)

Month	Jan 2022	Jul 2022	Oct 2022	Jan 2023	Jul 2023	Jul 2024
Forex reserves (US\$bn)	1,386	1,323	1,194	1,228	1,253	1,219

In September and October 2022, Japan spent over ¥9 trillion (US\$65bn) of its reserves intervening to support the yen — the largest yen-buying intervention in history. A further ¥9.8 trillion was spent in April and May 2024. While these interventions temporarily slowed the yen's slide, they did not reverse the trend until the BoJ finally raised rates in July 2024.

The weak yen has had mixed effects on Japan's current account. Japan's famed exporters — Toyota, Sony, Nintendo — have benefited from more competitive prices abroad and a translation boost when foreign earnings are converted to yen. Japan's tourism sector has boomed, with inbound visitor numbers exceeding pre-COVID levels in 2024. However, Japan's energy and food import bill has ballooned, and many of Japan's export 'champions' now produce much of their output overseas, reducing the boost to the trade balance from a weak yen.

Some economists argue Japan's problems are structural rather than cyclical. Japan's population is ageing rapidly — the working-age population has shrunk by over 12 million since 1995. Productivity growth has been sluggish (around 0.5% per year since 2000). Labour market rigidities mean wages adjust slowly, and 'lifetime employment' norms can discourage labour mobility. Critics of 'Abenomics' — the reform programme launched by the late PM Shinzo Abe in 2013 — argue that while the first two 'arrows' (loose monetary policy, fiscal expansion) were fired, the crucial 'third arrow' of structural reform was never delivered.

Table 2: Japan — selected indicators

Indicator	Japan	OECD average	G7 average
GDP growth (2015–2023, annual avg)	0.7%	1.9%	1.6%
Productivity (GDP per hour, US\$)	53.4	57.2	71.8
Old-age dependency ratio (65+ / working age, %)	50.3	31.7	37.4
Female labour force participation (%)	73.6	65.1	66.7
Public debt (% of GDP)	263.0	90.4	121.8
Corporate R&D (% of GDP)	2.6	1.8	2.2

In October 2024, a new Japanese government announced a ¥13 trillion fiscal support package to ease cost-of-living pressures and support small businesses. The BoJ signalled further gradual rate rises through 2025. But some commentators argue that after three decades of unconventional monetary policy, Japan needs deeper reform: immigration, greater female workforce participation, corporate governance reform, and a rethink of the energy mix to reduce import dependence. Whether the new administration can deliver where previous ones have failed remains to be seen.

6.

(a)

(i) Using the data in Table 1, calculate the percentage change in the yen price of LNG (liquefied natural gas) between 2020 and 2023. You are advised to show your working. **[2]**

(ii) 'The yen weakened from ¥103/US\$ in Jan 2021 to ¥150/US\$ in Oct 2022' (Figure 1). Using this information and the fact that crude oil is priced in US dollars on world markets, show how the yen depreciation affected the yen price of imported crude oil over this period, assuming the US dollar price of oil was constant. You are advised to show your working. **[2]**

(b) With reference to the data and using a cost and revenue diagram, show how the weak yen has resulted in higher prices, fewer sales and falling profits for Japanese firms that import commodities. **[6]**

7. Evaluate the extent to which Japan's economic difficulties in 2022–2024 were the result of external rather than internal factors. [8]

8. Discuss whether the Bank of Japan should have intervened in the ways described in the case (raising interest rates and/or direct currency market intervention) to strengthen the yen. **[10]**

9. With reference to the case study, discuss whether living standards in Japan are likely to be improved by a programme of deeper structural reform, including greater immigration, female workforce participation and corporate governance reform. **[12]**

END OF PAPER